Introduction to Computer Science

Lecture4

Software

OBJECTIVES AND INTENDED OUTCOMES

- By the end of this lesson the learner is expected to:
  - Understand the concept of software, program, programming, and programming languages.
  - Learn the types of software.
  - Learn the system software and application software.
  - Understand the concept of the software components,

Software

- **Software** non-physical units that drive the computer to perform tasks that are performed by the hardware components.

- The software is divided into two parts:
  - System software
    - Translation software
    - Operating Systems
  - Application software
    - user programs
    - Packages software

system software

- **system software** refers to the programs that manage the computer. divided into:

  - **Compiler Software**: programs that convert a program written in programming language into an executable program written in machine language

  - **Operating System:** a set of basic programs that acts as an interface between the computer user and computer hardware, and controls the execution of programs.
    - Ex: Mac , Windows, Linux…

system software Cont..

- The common management tasks that an operating system perform include:
  - File Management
  - User Management
  - Memory Management
  - Process Management
  - Device Management
  - Security Management
  - User Interface Management

Application software

- **Application Software:** Programs designed to perform a specific task for users.

- Divided into two types:
  - **User Programs**: Each user can write their own programs to solve a specific issue using one of the programming languages.
  - **Packages software:** Developed by software production companies.

programs

- **Programs** refer to instructions that are directed to a computer for the purpose of performing a specific task, and computers need programs to perform each task they perform.

- Writing a program is called **programming**, and whoever does it is called a **programmer**.

- **Programming** It is preparing a plan to process a task and get the desired results

- **A programming language** is a system of notation for writing computer programs. programming languages are text-based formal languages.